

F1 Exponentials and graphs

Name: _____

Class: _____

Date: _____

Time: **334 minutes**

Marks: **279 marks**

Comments:

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Q1.

Trees in a forest may be affected by one of two types of fungal disease, but not by both.

The number of trees affected by disease A, n_A , can be modelled by the formula

$$n_A = ae^{0.1t}$$

where t is the time in years after 1 January 2017.

The number of trees affected by disease B, n_B , can be modelled by the formula

$$n_B = be^{0.2t}$$

On 1 January 2017 a **total** of 290 trees were affected by a fungal disease.

On 1 January 2018 a **total** of 331 trees were affected by a fungal disease.

- (a) Show that $b = 90$, to the nearest integer, and find the value of a . (3)
- (b) Estimate the total number of trees that will be affected by a fungal disease on 1 January 2020. (1)
- (c) Find the year in which the number of trees affected by disease B will first exceed the number affected by disease A. (3)
- (d) Comment on the long-term accuracy of the model. (1)

(Total 8 marks)

Q2.

The graph of $y = 5^x$ is transformed by a stretch in the y -direction, scale factor 5

State the equation of the transformed graph.

Circle your answer.

$$y = 5 \times 5^x$$

$$y = 5^{\frac{x}{5}}$$

$$y = \frac{1}{5} \times 5^x$$

$$y = 5^{5x}$$

(Total 1 mark)

Q3.

A scientist is researching the effects of caffeine. She models the mass of caffeine in the body using

$$m = m_0 e^{-kt}$$

where m_0 milligrams is the initial mass of caffeine in the body and m milligrams is the mass of caffeine in the body after t hours.

On average, it takes 5.7 hours for the mass of caffeine in the body to halve.

One cup of strong coffee contains 200 mg of caffeine.

- (a) The scientist drinks two strong cups of coffee at 8 am. Use the model to estimate the mass of caffeine in the scientist's body at midday.

(4)

- (b) The scientist wants the mass of caffeine in her body to stay below 480 mg

Use the model to find the earliest time that she could drink another cup of strong coffee.

Give your answer to the nearest minute.

(3)

- (c) State a reason why the mass of caffeine remaining in the scientist's body predicted by the model may not be accurate.

(1)

(Total 8 marks)

Q4.

A curve has equation $y = e^{2x}$

Find the coordinates of the point on the curve where the gradient of the curve is $\frac{1}{2}$
Give your answer in an exact form.

(Total 5 marks)

Q5.

David has been investigating the population of rabbits on an island during a three-year period.

Based on data that he has collected, David decides to model the population of rabbits, R , by the formula

$$R = 50e^{0.5t}$$

where t is the time in years after 1 January 2016.

- (a) Using David's model:

- (i) state the population of rabbits on the island on 1 January 2016;

(1)

- (ii) predict the population of rabbits on 1 January 2021.

(1)

- (b) Use David's model to find the value of t when $R = 150$, giving your answer to three significant figures.

(2)

- (c) Give **one** reason why David's model may **not** be appropriate.

(1)

- (d) On the same island, the population of crickets, C , can be modelled by the formula

$$C = 1000e^{0.1t}$$

where t is the time in years after 1 January 2016.

Using the two models, find the year during which the population of rabbits first exceeds the population of crickets.

(3)

(Total 8 marks)

Q6.

The function f is defined by

$$f(x) = 4 + 3^{-x}, x \in \mathbb{R}$$

- (a) Using set notation, state the range of f

(2)

- (b) The inverse of f is f^{-1}

- (i) Using set notation, state the domain of f^{-1}

(1)

- (ii) Find an expression for $f^{-1}(x)$

(3)

- (c) The function g is defined by

$$g(x) = 5 - \sqrt{x}, (x \in \mathbb{R} : x > 0)$$

- (i) Find an expression for $gf(x)$

(1)

- (ii) Solve the equation $gf(x) = 2$, giving your answer in an exact form.

(3)

(Total 10 marks)

Q7.

- (a) Three consecutive terms in an arithmetic sequence are $3e^{-p}$, 5 , $3e^p$

Find the possible values of p . Give your answers in an exact form.

(6)

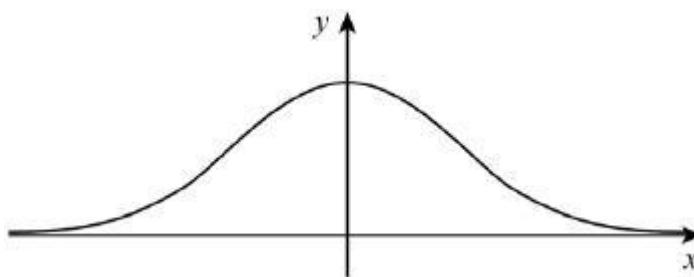
- (b) Prove that there is no possible value of q for which $3e^{-q}$, 5 , $3e^q$ are consecutive terms of a geometric sequence.

(4)

(Total 10 marks)

Q8.

The diagram shows part of the graph of $y = e^{-x^2}$

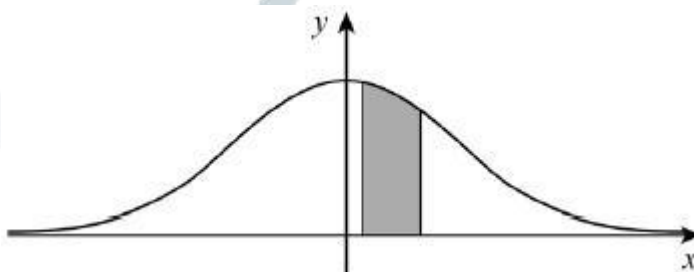


The graph is formed from two convex sections, where the gradient is increasing, and one concave section, where the gradient is decreasing.

- (a) Find the values of x for which the graph is concave.

(4)

- (b) The finite region bounded by the x -axis and the lines $x = 0.1$ and $x = 0.5$ is shaded.



Use the trapezium rule, with 4 strips, to find an estimate for $\int_{0.1}^{0.5} e^{-x^2} dx$

Give your estimate to four decimal places.

(3)

- (c) Explain with reference to your answer in part (a), why the answer you found in part (b) is an underestimate.

(2)

- (d) By considering the area of a rectangle, and using your answer to part (b), prove that the shaded area is 0.4 correct to 1 decimal place.

(3)

(Total 12 marks)

Q9.

- (a) Describe a geometrical transformation that maps the graph of $y = 4^x$ onto the graph of $y = 3 \times 4^x$. (2)
- (b) Sketch the curve with equation $y = 3 \times 4^x$, indicating the value of the intercept on the y -axis. (2)
- (a) The curve with equation $y = 4^{-x}$ intersects the curve $y = 3 \times 4^x$ at the point P . Use logarithms to find the x -coordinate of P , giving your answer to three significant figures. (5)
- (Total 9 marks)**

Q10.

- (a) Sketch the graph of $y = 9^x$, indicating the value of the intercept on the y -axis. (2)
- (b) Use logarithms to solve the equation $9^x = 15$, giving your value of x to three significant figures. (2)
- (c) The curve $y = 9^x$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)
- (Total 5 marks)**

Q11.

- (a) Sketch the graph of $y = \frac{1}{2^x}$, indicating the value of the intercept on the y -axis. (2)
- (b) Use logarithms to solve the equation $\frac{1}{2^x} = \frac{5}{4}$, giving your answer to three significant figures. (3)
- (c) Given that

$$\log_a(b^2) + 3 \log_a y = 3 + 2 \log_a \left(\frac{y}{a} \right)$$

express y in terms of a and b .

Give your answer in a form not involving logarithms.

(5)
(Total 10 marks)

Q12.

A scientist is testing models for the growth and decay of colonies of bacteria.

For a particular colony, which is growing, the model is $P = Ae^{\frac{1}{8}t}$, where P is the number of bacteria after a time t minutes and A is a constant.

- (a) This growing colony consists initially of 500 bacteria. Calculate the number of bacteria, according to the model, after one hour. Give your answer to the nearest thousand.

(2)

- (b) For a second colony, which is decaying, the model is $Q = 500\,000e^{-\frac{1}{8}t}$, where Q is the number of bacteria after a time t minutes.

Initially, the growing colony has 500 bacteria and, at the same time, the decaying colony has 500 000 bacteria.

- (i) Find the time at which the populations of the two colonies will be equal, giving your answer to the nearest 0.1 of a minute.

(3)

- (ii) The population of the growing colony will exceed that of the decaying colony by 45 000 bacteria at time T minutes.

Show that

$$\left(e^{\frac{1}{8}T}\right)^2 - 90e^{\frac{1}{8}T} - 1000 = 0$$

and hence find the value of T , giving your answer to one decimal place.

(4)

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(Total 9 marks)

Q13.

- (a) Sketch the curve with equation $y = 7^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) The curve C_1 has equation $y = 7^x$.
The curve C_2 has equation $y = 7^{2x} - 12$.

- (i) By forming and solving a quadratic equation, prove that the curves C_1 and C_2 intersect at exactly one point. State the y -coordinate of this point.

(4)

- (ii) Use logarithms to find the x -coordinate of the point of intersection of C_1 and C_2 , giving your answer to three significant figures.

(2)
(Total 8 marks)

Q14.

(a) Write down the values of p , q and r given that:

(i) $8 = 2^p$; (1)

(ii) $\frac{1}{8} = 2^q$; (1)

(iii) $\sqrt{2} = 2^r$. (1)

(b) Find the value of x for which $\sqrt{2} \times 2^x = \frac{1}{8}$.

(2)
(Total 5 marks)

Q15.

A model for the radioactive decay of a form of iodine is given by

$$m = m_0 2^{-\frac{1}{8}t}$$

The mass of the iodine after t days is m grams. Its initial mass is m_0 grams.

(a) Use the given model to find the mass that remains after 10 grams of this form of iodine have decayed for 14 days, giving your answer to the nearest gram. (2)

(b) A mass of m_0 grams of this form of iodine decays to $\frac{m_0}{16}$ grams in d days.

Find the value of d . (2)

(c) After n days, a mass of this form of iodine has decayed to less than 1% of its initial mass.

Find the minimum integer value of n . (3)

(3)
(Total 7 marks)

Q16.

- (a) Sketch the curve with equation $y = 4^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) Describe the geometrical transformation that maps the graph of $y = 4^x$ onto the graph of $y = 4^x - 5$.

(2)

- (c) (i) Use the substitution $Y = 2^x$ to show that the equation $4^x - 2^{x+2} - 5 = 0$ can be written as $Y^2 - 4Y - 5 = 0$.

(2)

- (ii) Hence show that the equation $4^x - 2^{x+2} - 5 = 0$ has only one real solution.

Use logarithms to find this solution, giving your answer to three decimal places.

(4)

(Total 10 marks)

Q17.

- (a) Sketch the curve with equation $y = 2^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 2^x \, dx$, giving your answer to three significant figures.

(4)

- (ii) State how you could obtain a better approximation to the value of the integral using the trapezium rule.

(1)

- (c) Describe a geometrical transformation that maps the graph of $y = 2^x$ onto the graph of $y = 2^{x+7} + 3$.

(3)

- (d) The curve $y = 2^{x+k} + 3$ intersects the y -axis at the point $A(0, 8)$.

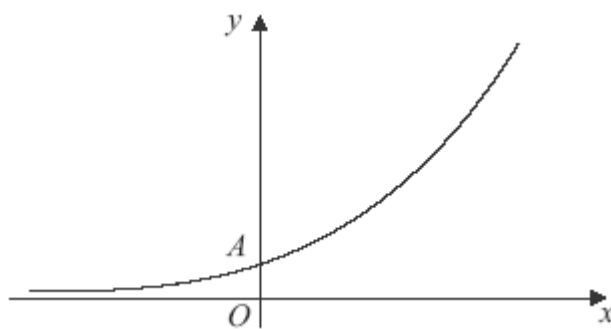
Show that $k = \log_m n$, where m and n are integers.

(2)

(Total 12 marks)

Q18.

The diagram shows a sketch of the curve $y = 2^{4x}$.



The curve intersects the y -axis at the point A .

- (a) Find the value of the y -coordinate of A . (1)

- (b) Use the trapezium rule with six ordinates (five strips) to find an approximate value for $\int_0^1 2^{4x} \, dx$, giving your answer to two decimal places. (4)

- (c) Describe the geometrical transformation that maps the graph of $y = 2^{4x}$ onto the graph of $y = 2^{4x-3}$. (2)

- (d) The curve $y = 2^{4x}$ is translated by the vector $\begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix}$ to give the curve $y = g(x)$.
The curve $y = g(x)$ crosses the x -axis at the point Q . Find the x -coordinate of Q . (4)

- (e) (i) Given that $\log_a k = 3 \log_a 2 + \log_a 5 - \log_a 4$
show that $k = 10$. (3)

- (ii) The line $y = \frac{5}{4}$ crosses the curve $y = 2^{4x-3}$ at the point P . Show that the x -coordinate of P is $\frac{1}{4 \log_{10} 2}$. (3)

(Total 17 marks)

Q19.

- (a) Sketch the graph of $y = 3^x$, stating the coordinates of the point where the graph crosses the y -axis. (2)

- (b) Describe a single geometrical transformation that maps the graph of $y = 3^x$:

- (i) onto the graph of $y = 3^{2x}$; (2)

- (ii) onto the graph of $y = 3^{x+1}$. (2)

- (c) (i) Using the substitution $Y = 3^x$, show that the equation

$$9^x - 3^{x+1} + 2 = 0$$

can be written as

$$(Y - 1)(Y - 2) = 0$$

- (ii) Hence show that the equation $9^x - 3^{x+1} + 2 = 0$ has a solution $x = 0$ and, by using logarithms, find the other solution, giving your answer to four decimal places. (4)

(Total 12 marks)

Q20.

The diagram shows a sketch of the curve with equation $y = 6^x$.



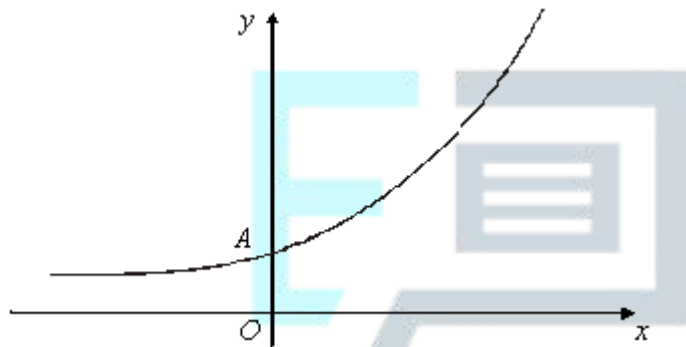
- (a) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 6^x dx$, giving your answer to three significant figures. (4)

- (ii) Explain, with the aid of a diagram, whether your approximate value will be an overestimate or an underestimate of the true value of $\int_0^2 6^x dx$. (2)

- (b) (i) Describe a single geometrical transformation that maps the graph of $y = 6^x$ onto the graph of $y = 6^{3x}$. (2)
- (ii) The line $y = 84$ intersects the curve $y = 6^{3x}$ at the point A . By using logarithms, find the x -coordinate of A , giving your answer to three decimal places. (4)
- (c) The graph of $y = 6^x$ is translated by $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ to give the graph of the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (2)
- (Total 14 marks)

Q21.

The diagram shows a sketch of the curve with equation $y = 3(2^x + 1)$.



The curve $y = 3(2^x + 1)$ intersects the y -axis at the point A .

- (a) Find the y -coordinate of the point A . (2)
- (b) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^6 3(2^x + 1) dx$ (4)
- (c) The line $y = 21$ intersects the curve $y = 3(2^x + 1)$ at the point P .
- (i) Show that the x -coordinate of P satisfies the equation $2^x = 6$ (1)
- (ii) Use logarithms to find the x -coordinate of P , giving your answer to three significant figures. (3)

(Total 10 marks)

Q22.

A biologist is researching the growth of a certain species of hamster. She proposes that the length, x cm, of a hamster t days after its birth is given by

$$x = 15 - 12e^{-\frac{t}{14}}$$

(a) Use this model to find:

(i) the length of a hamster when it is born;

(1)

(ii) the length of a hamster after 14 days, giving your answer to three significant figures.

(2)

(b) (i) Show that the time for a hamster to grow to 10 cm in length is given by

$$t = 14 \ln \left(\frac{a}{b} \right),$$

where a and b are integers.

(3)

(ii) Find this time to the nearest day.

(1)

(c) (i) Show that

$$\frac{dx}{dt} = \frac{1}{14}(15 - x)$$

(3)

(ii) Find the rate of growth of the hamster, in cm per day, when its length is 8 cm.

(1)

(Total 11 marks)

Q23.

(a) Use logarithms to solve the equation $0.8^x = 0.05$, giving your answer to three decimal places.

(3)

(b) An infinite geometric series has common ratio r . The sum to infinity of the series is five times the first term of the series.

(i) Show that $r = 0.8$.

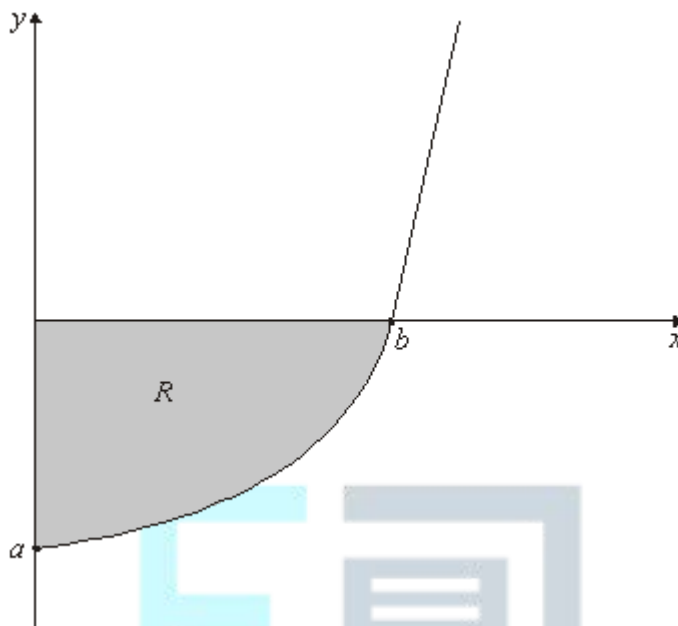
(3)

(ii) Given that the first term of the series is 20, find the least value of n such that the n th term of the series is less than 1.

(3)
(Total 9 marks)

Q24.

The diagram shows part of the graph of $y = e^{2x} - 9$. The graph cuts the coordinate axes at $(0, a)$ and $(b, 0)$.



- (a) State the value of a , and show that $b = \ln 3$.

(3)

- (b) Show that $y^2 = e^{4x} - 18e^{2x} + 81$.

(1)

(Total 4 marks)

Q25.

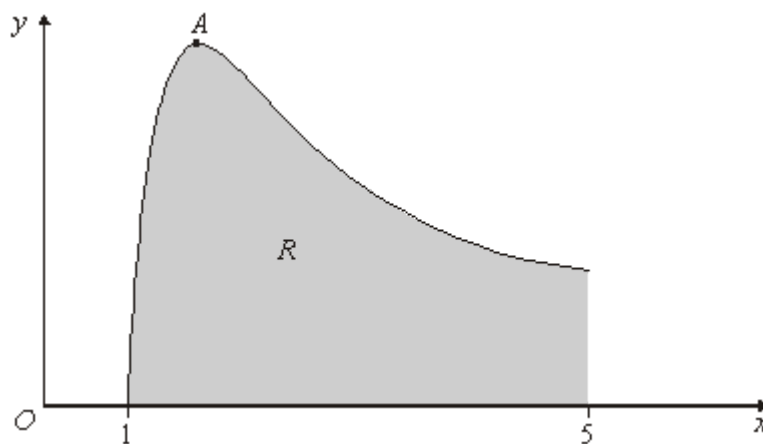
- (a) Given that $y = x^{-2} \ln x$, show that $\frac{dy}{dx} = \frac{1 - 2 \ln x}{x^3}$.

(4)

- (b) Using integration by parts, find $\int x^{-2} \ln x \, dx$.

(4)

- (c) The sketch shows the graph of $y = x^{-2} \ln x$.



- (i) Using the answer to part (a), find, in terms of e , the x -coordinate of the stationary point A . (2)
- (ii) The region R is bounded by the curve, the x -axis and the line $x = 5$. Using your answer to part (b), show that the area of R is

$$\frac{1}{5} (4 - \ln 5)$$

(3)
(Total 13 marks)

Q26.

On 1 January 1900, a sculpture was valued at £80.

When the sculpture was sold on 1 January 1956, its value was £5000.

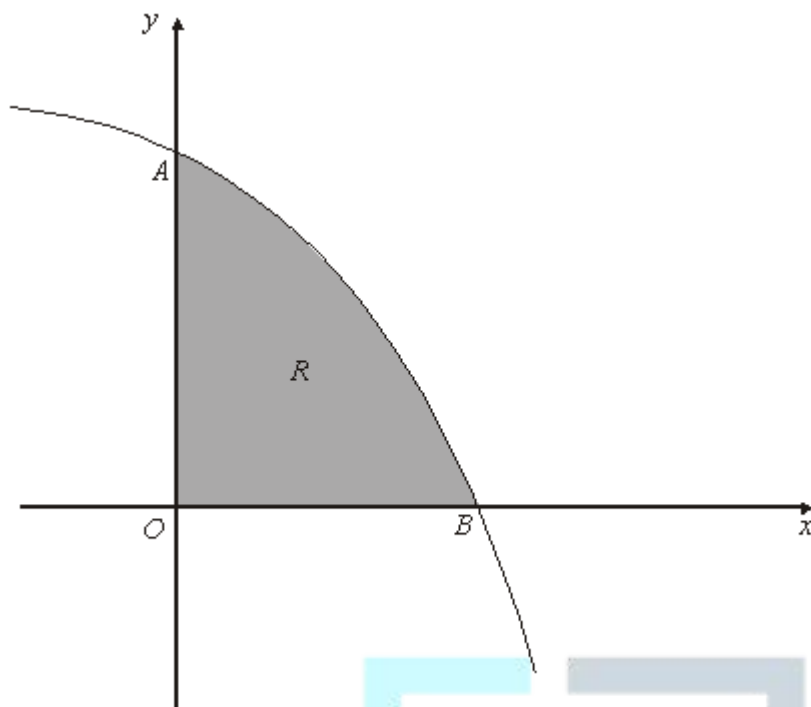
The value, £ V , of the sculpture is modelled by the formula $V = A k^t$, where t is the time in years since 1 January 1900 and A and k are constants.

- (a) Write down the value of A . (1)
- (b) Show that $k \approx 1.07664$. (3)
- (c) Use this model to:
- (i) show that the value of the sculpture on 1 January 2006 will be greater than £200 000; (2)
- (ii) find the year in which the value of the sculpture will first exceed £800 000. (3)

(Total 9 marks)

Q27.

The diagram shows a sketch of the curve with equation $y = 27 - 3^x$.



The curve $y = 27 - 3^x$ intersects the y -axis at the point A and the x -axis at the point B .

- (a) (i) Find the y -coordinate of point A . (2)
- (ii) Verify that the x -coordinate of point B is 3. (1)
- (b) The region, R , bounded by the curve $y = 27 - 3^x$ and the coordinate axes is shaded. Use the trapezium rule with four ordinates (three strips) to find an approximate value for the area of R . (4)
- (c) (i) Use logarithms to solve the equation $3^x = 13$, giving your answer to four decimal places. (3)
- (ii) The line $y = k$ intersects the curve $y = 27 - 3^x$ at the point where $3^x = 13$.
Find the value of k . (1)
- (d) (i) Describe the single geometrical transformation by which the curve with equation $y = -3^x$ can be obtained **from** the curve $y = 27 - 3^x$. (2)

- (ii) Sketch the curve $y = -3^x$.

(2)

(Total 15 marks)

Q28.

A function f is defined by $f(x) = 2e^{3x} - 1$ for all real values of x .

- (a) Find the range of f .

(2)

- (b) Show that $f^{-1}(x) = \frac{1}{3} \ln \left(\frac{x+1}{2} \right)$.

(3)

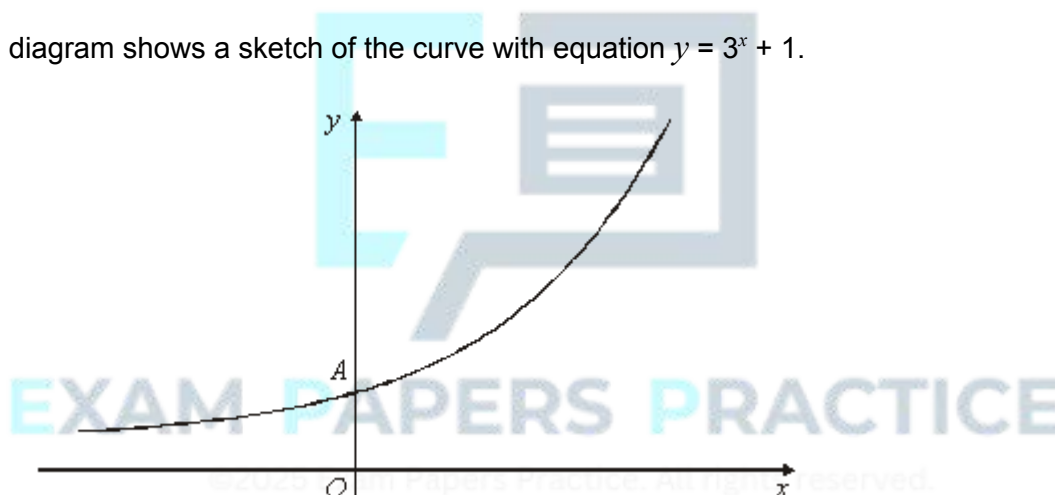
- (c) Find the gradient of the curve $y = f^{-1}(x)$ when $x = 0$.

(4)

(Total 9 marks)

Q29.

The diagram shows a sketch of the curve with equation $y = 3^x + 1$.



The curve intersects the y -axis at the point A .

- (a) Write down the y -coordinate of point A .

(1)

- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximation for

$$\int_0^1 (3^x + 1) dx$$

, giving your answer to three significant figures.

(4)

- (ii) By considering the graph of $y = 3^x + 1$, explain with the aid of a diagram whether your approximation will be an overestimate or an underestimate of the true value of

$$\int_0^1 (3^x + 1) dx$$

(2)

- (c) The line $y = 5$ intersects the curve $y = 3^x + 1$ at the point P . By solving a suitable equation, find the x -coordinate of the point P . Give your answer to four decimal places.

(4)

- (d) The curve $y = 3^x + 1$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$.

(1)

(Total 12 marks)

Q30.

- (a) Solve the equation $2e^x = 5$, giving your answer as an exact natural logarithm.

(2)

- (b) (i) By substituting $y = e^x$, show that the equation $2e^x + 5e^{-x} = 7$ can be written as

$$2y^2 - 7y + 5 = 0$$

(2)

- (ii) Hence solve the equation $2e^x + 5e^{-x} = 7$, giving your answers as exact values of x .

(3)

(Total 7 marks)

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